Ayush Dhangar

[Email] | [Phone] | [LinkedIn Profile] | [GitHub Profile] | [Portfolio (if applicable)]

EDUCATION

University of North Carolina at Chapel Hill | Chapel Hill, NC **May 2027**

Master of Science in Computer Science

- Relevant Coursework: Machine Learning, Distributed Systems, Algorithms, Cloud Computing
- Research: Developing scalable deep learning models for real-time fraud detection in financial transactions (advised by Dr. [Advisor's Name])
- Thesis: "**Enhancing Anomaly Detection in Large-Scale Systems Using AI**"

North Carolina State University | Raleigh, NC **May 2023**

Bachelor of Science in Computer Science and Environmental Science

Honors: Dean's List (all semesters), Magna cum Laude

INDUSTRY EXPERIENCE

IBM – Research Triangle Park, NC **June 2023 – July 2026**

Software Engineer

- Developed Java microservices deployed with **Docker and Kubernetes** to support scalable enterprise applications
- Optimized **SQL** queries and backend services, reducing application response times by up to **30%**
- Implemented **REST APIs** and integrated cloud-hosted services to improve system interoperability
- Collaborated in **Agile** sprints, participating in code reviews, technical design discussions, and production deployments
- Architected automated testing and real-time monitoring solutions that reduced deployment failure rates by **25%**; integrated AI-assisted development tools to accelerate the CI/CD pipeline and boost code coverage to **92%**

Fidelity Investments – Durham, NC **May 2022 – August 2022**

Software Engineer Intern

- Engineered a **React and Node.js dashboard** for real-time stock analysis, improving data refresh rates by **50%**
- Created RESTful APIs using **Java Spring Boot** to integrate investment analytics with customer dashboards
- Developed SQL queries to optimize **financial data retrieval**, reducing query execution time by 35%

RESEARCH EXPERIENCE

UNC-Chapel Hill – Department of Computer Science (**Cybersecurity Lab**) **January 2026 – Present**

Graduate Research Assistant

- Investigate the impact of **adversarial attacks on neural networks**, proposing a **defensive training approach** that improved model robustness by **18%**
- Conduct **data preprocessing and feature engineering** on large-scale cybersecurity datasets using **Spark and Pandas**
- Developed reproducible MLOps pipelines using Python and TensorFlow to automate the training, evaluation, and benchmarking of **10+** defensive models, reducing experimentation cycle time by **40%**
- Presented findings at the **IEEE Workshop on Machine Learning for Security**

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Frameworks: React, Node.js, Spring Boot, REST APIs

Cloud & DevOps: AWS, Docker, Kubernetes, Git, PostgreSQL

Methodologies: Agile, Test-Driven Development, CI/CD